

Reader Supplement to *Generic Identifiability and Directed Containment for Strongly Tree-Child Level-2 Networks under the Kimura Two-Parameter Model*

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Abstract

This supplement is a reader-oriented map from the mathematical proof to the exact finite certificate package. It separates hand proofs from the bounded computer-assisted residue, defines the primitive completion grammar and raw relation object, records all censuses and restoration/probe invariants, gives representative certificate schemas and the complete weak-class witness, and lists the minimal artifacts and commands needed to replay each theorem layer. The frozen theorem release remains unchanged; this document is a derived submission guide.

Contents

1	How to read the proof	2
2	Convention crosswalk	3
3	Primitive graph grammar	3
3.1	Core segments and minimum repairs	3
3.2	Completion grammar and the numbers 831, 1983, and 4155	4
3.3	Raw relation schema	4
4	Certificate decision tree	4
5	Finite census and current compression boundary	5
5.1	Authoritative raw partitions	5
5.2	Audited proof-compression table	5
5.3	Restoration forests	6
6	Coherent probes and arbitrary subdivision words	7

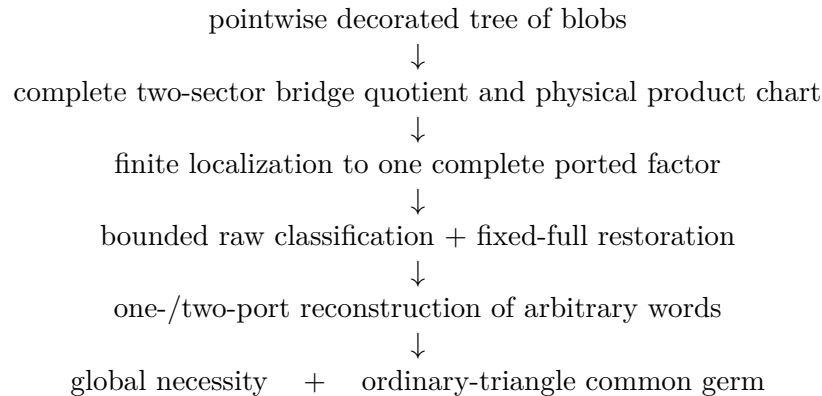
7	Representative exact calculations	8
7.1	Tree–sunlet factorization	8
7.2	Bridge normalizer	8
7.3	Certificate schema for a direct polynomial	8
8	Full weak-class certificate	8
9	Theorem-to-artifact crosswalk	9
10	Frozen hash anchors	12
11	Replay protocol	12
12	Mutation coverage	13
13	Recent-source verification note	13
14	Pending submission metadata and licenses	14

1 How to read the proof

The main article proves a global theorem from three qualitatively different types of input.

- P1. Analytic and algebraic proofs.** These include the physical K2P domain, subdivision and rerooting, pointwise quartet and whole-map $\mathcal{T}_i$ signs, the two-sector bridge fibre, physical local products, marginal submersions, localization, genericity, continuous-time gluing, and the weak-class construction.
- P2. Topology-only finite reduction.** The cycle/theta primitive theorem and minimum-repair grammar are graph-theoretic. They reduce an arbitrary strong level-2 factor to a finite rigid support plus ordered labelled words on finitely many directed segments.
- P3. Exact bounded computation.** Raw four- and five-port directions, rank exclusions, direct polynomial separators, restoration children, and coherent probes are exhausted by exact graph and rational/symbolic arithmetic. This layer has independent regeneration and mutations, but is not presented as a complete short hand classification.

The logical dependency is



The quartet and $\mathcal{T}_i$ inequalities are pointwise. The polynomial and rank certificates are generic non-containment statements. Restoration applies only inside one assumed fixed full relation. Keeping these quantifiers separate is essential.

2 Convention crosswalk

All graphs use the following fixed convention.

- Rooted presentations are binary directed acyclic graphs; the root is the lowest stable ancestor.
- Only arcs entering reticulations retain arrowheads.
- The root is deleted once and its two incident arcs merge into one mixed edge. No later parallel-edge or degree-two cleanup is part of formation of the full standard semi-directed graph.
- Restriction/display operations may subsequently prune dead ends and suppress unlabelled bivalent vertices; they do not change the admissible-rooting universe of the original graph.
- “Strongly tree-child” quantifies over every admissible rooting; “weakly tree-child” requires at least one tree-child rooting.
- Isomorphisms preserve labels, ordinary edges, arrowheads, and vertex roles. Ordinary triangle equivalence forgets only the reticulation choice on one three-cycle and keeps its three external arms fixed.

The K2P character order is $(0, C, G, T)$, with equal sector $\{C, T\}$ and singleton sector $\{G\}$. Every edge has Fourier vector $(1, s, g, s)$ in

$$\mathcal{D}_+ = \{0 < s < 1, 0 < g < 1, g > 2s - 1\}.$$

The strict continuous-time subdomain is

$$\mathcal{D}_{CT} = \{0 < s < 1, s^2 < g < 1\}.$$

No mixed-sign stochastic component is part of the theorem.

3 Primitive graph grammar

3.1 Core segments and minimum repairs

After removing external port arms and suppressing ordinary non-event degree-two vertices, a nontrivial strong level-2 blob is a cycle or one of four directed theta event placements. Segment indices in the last column below refer to the left-to-right segment order.

core	directed segments	minimum repair sets
cycle	$S \rightarrow X, S \rightarrow X$	$\{0\}, \{1\}$
θ_0	$S \rightarrow U, S \rightarrow V, U \rightarrow X, V \rightarrow X, U \rightarrow V$	$\{2, 3\}, \{3, 4\}$
θ_1	$S \rightarrow U, S \rightarrow X, V \rightarrow X, U \rightarrow V, U \rightarrow V$	$\{2, 3\}, \{2, 4\}$
θ_2	$S \rightarrow U, S \rightarrow V, U \rightarrow X_0, V \rightarrow X_0, U \rightarrow X_1, V \rightarrow X_1$	$\{2, 3\}, \{2, 5\}, \{3, 4\}, \{4, 5\}$
θ_3	$S \rightarrow U, S \rightarrow X_0, V \rightarrow X_0, U \rightarrow X_1, V \rightarrow X_1, U \rightarrow V$	$\{2\}, \{4\}$

An occupied segment contains at least one port-bearing subdivision. The minimum repairs are the minimal occupancy sets satisfying the no-omnian and simple-graph requirements. A complete factor additionally retains one child port below every path-sink reticulation. Arbitrary factors arise by replacing each occupied segment by an ordered word of distinct labelled ports.

3.2 Completion grammar and the numbers 831, 1983, and 4155

For each core the compiler enumerates:

- (i) every allowed subset of path-sink child roles;
- (ii) every weak composition of the remaining labelled port roles among the directed segments;
- (iii) every minimum-repair completion;
- (iv) the selected-incoming mode, in which a real incoming boundary remains a port, and the marginalized-incoming mode, in which that role has been omitted; and
- (v) every physical port permutation.

The directed completion grammar yields 831 four-port target descriptors in selected-incoming mode and 1,983 in marginalized-incoming mode. Five-port targets number 1,983 and 4,155 in the corresponding modes. These numbers count graph-derived target descriptors before source multiplication and port permutations. Thus

$$6(831 + 1983)4! = 405,216, \quad 4(1983 + 4155)5! = 2,946,240.$$

3.3 Raw relation schema

A raw directional relation is not an unlabelled topology hash. Its immutable identity includes:

- source primitive and exact repaired support;
- target primitive, completion descriptor, and selected graph;
- source and target incoming roles;
- physical port permutation and port match;
- omitted source and target roles;
- source-to-target direction;
- boundary incidence transport; and
- exact graph fingerprints.

Only after every raw ID is generated exactly once are presentations grouped into canonical terminal or restoration classes.

4 Certificate decision tree

The classifier uses proof types in a fixed precedence order.

type	mathematical implication	exact stored evidence
Displayed quartet	physical images are disjoint point-wise	distinct complete switching-split sets, selected witness split, and the appropriate leaf-permuted F or G polynomial.
Whole-map $\mathcal{T}_i$	physical images are disjoint point-wise	coordinate triple and orientation, exact whole-map pullbacks, and a strict Bernstein/sign certificate for the factorized pullback.
Directed rank	no full-dimensional source germ enters the target	nonzero exact source minor, target lower minor, symbolic target rank upper certificate, and strict source-rank $>$ target-rank comparison.

type	mathematical implication	exact stored evidence
Polynomial separator	generic source-to-target containment is impossible	sparse integer polynomial $P(q)$, coefficientwise proof $P \circ \Phi_T = 0$, nonzero source pullback, and strict rational source witness.
Restoration parent	deferred fixed-full obligation	parent relation, each actual physical child attachment, exact parent-child transport restriction, and a terminal certificate at every leaf.
Isomorphism	physical parameter transport exists	exact labelled mixed-incidence isomorphism and parameter/port transport.
Ordinary triangle	one common full-dimensional regular germ exists	stored triangle edge set and coherent transport, combined with the exact rank-nine analytic construction in the article.

The target rank upper certificate is symbolic; a one-point Jacobian rank is never used as an upper bound. Polynomial witnesses are checked on the graph-generated K2P maps rather than accepted by a topology identifier.

5 Finite census and current compression boundary

5.1 Authoritative raw partitions

category	four-port theta	five-port θ_2
displayed quartet	360,408	2,942,592
whole-map $\mathcal{T}_i$	16,974	2,528
directed exact rank	23,822	800
direct terminal / quadratic	1,472	240
restoration member / isomorphism	2,540	80
total	405,216	2,946,240

The 1,472 four-port terminal presentations form 934 classes:

839 quadratics + 36 direct polynomials + 4 hard bindings + 20 isomorphisms + 35 triangles.

The 36 direct polynomials have three certified source-family tags: 22 θ_0 quintic port-orbit obstructions, 12 lower-theta quartics, and two θ_3 cubics. The 839 quadratic classes have eight literal certificate bodies. These repetitions make further theorem-template compression plausible, but equality of certificate bodies is not by itself a proof that all graph transports belong to one legal orbit.

5.2 Audited proof-compression table

The bounded compression pass has status PC-PARTIAL. It changes no theorem statement and discards no exact ledger. In particular, literal equality of polynomial bodies is not treated as a graph orbit, and the 297 restoration fingerprints are descriptive archetypes rather than a proved transport quotient.

layer	frozen units	compact units	exact residue retained
four-port quadratics	839	8	every coordinate transport
five-port θ_2 quadratics	96	4	every coordinate transport
three-port cycle quadratics	54	6	every coordinate transport
restoration quadratics	6	5	every coordinate transport
direct-36 high degree	36	27	all directional bindings; 3 certified proposition families, 5 stored base formulas
rank upper certificates	4,379	3,515+75	universal mechanism plus 75 exceptional representatives covering 864 descriptors
restoration parents	997	297	997 assignments, 2,540 roots, 36,824 edges, and 16 algebra-transport classes

The direct-36 row must not be read as three literal polynomials. The quintic, quartic, and cubic proposition families have 27 direction-specific bodies across the 36 records. The quadratic rows group only literal coordinate bodies inside their original coordinate convention or transports already certified by a frozen orbit package.

layer	uniform statement	load-bearing finite premise
primitive completions	one binomial-sum formula for all selected/marginalized cases	all labelled directed completion records; totals 405,216, 2,946,240, and 13,440
restoration	four terminal mechanisms and one depth-two continuation mechanism	the exact 997-parent assignment and every physical parent-child transport
probe reconstruction	one-/two-port coherence implies arbitrary attachment-word recovery	176 anchors, 29,964 one-port rows, 544,571 two-port rows, 67,741 transports, and 4,379 restrictions

The machine-readable theorem-to-artifact crosswalk is `proof_compression_submission/crosswalk/THEOREM_ARTIFACT_CROSSWALK.json`. It records, for each theorem layer, exact file hashes and declared schemas, producer, replay, mutation, and environment bindings. No byte-bound end-to-end quick or full runtime is present in the frozen release, so those times are marked unknown rather than inferred by summing noncomparable component observations.

5.3 Restoration forests

The corrected four-port restoration forest has 997 canonical parents and 2,540 physical member roots. Its first layer is

quartet separator	35,758	
whole-map $\mathcal{T}_i$	606	
quadratic	148	(36,568 first children).
transported quartic	24	
continuation	32	

The 32 continuations have 256 second children: 248 quartet and eight $\mathcal{T}_i$. Hence there are 36,824 edges, 36,792 exact separator leaves, maximum depth two, no cycle, no missing child, and no unresolved record. Every child transport restricts to its parent's boundary transport.

The five-port θ_2 dummy forest has 56 roots and 864 one-parent descendants. Its 832 leaves are 760 quartet exclusions and 72 labelled isomorphisms. The cycle layer has 5,964 restoration roots and 536,364 physical completions, of which 535,920 are quartet-separated, 300 are $\mathcal{T}_i$ -separated, 132 have exact quadratics, and 12 are isomorphic.

The logical implication is always fixed-full. An assumed full relation fixes the omitted physical label, source edge, target edge, role, and transport. That actual completion is one child in the forest. Source marginal openness then turns its separator into a contradiction. No abstract parent relation is lifted, and no target deletion map is inverted.

6 Coherent probes and arbitrary subdivision words

The equality-anchor input contains 176 physical relations:

origin	ports	records
four-port direct terminals	4	26
restored four-port terminals	5	17
θ_2 direct anchors	5	24
θ_2 one-role restorations	6	40
θ_2 two-role paths	7	32
cycle direct anchors	3	24
cycle restored anchors	4	12
ordinary tree identity	3	1
total		176

There are 143 isomorphism and 33 triangle anchors.

If a rooted binary factor has k boundary leaves and r reticulations, degree counting gives $t = k + r - 2$ nonroot tree vertices and $2k + 3r - 2$ rooted arcs. Root suppression merges two arcs, leaving exactly

$$2k + 3r - 3$$

physical attachment sites. The probe input contains every site: the root-suppressed incoming segment, every other pendant arm, both incoming edges at every reticulation, and every remaining unheaded core edge. The two artificial root halves are identified only after exact semi-directed isomorphism.

Across all anchors there are 2,206 source sites and 2,206 target sites. The first-probe universe is the per-anchor Cartesian product and contains 29,964 directions. Its census is

$$27,758 \text{ quartet} + 99 \mathcal{T}_i + 1,915 \text{ isomorphism} + 192 \text{ triangle}.$$

The 2,107 equality survivors generate every two-port parent exactly once. The second-probe universe has 544,571 directions:

$$511,266 \text{ quartet} + 576 \mathcal{T}_i + 30,969 \text{ isomorphism} + 1,760 \text{ triangle}.$$

One-port restrictions identify the directed core segment containing each omitted boundary. Two-port restrictions determine the order of every pair on one segment. Since the comparisons come from a genuine word, they are transitive and recover a unique total order. The exact child transports agree on overlaps and restrict to their parents. All 33 triangle anchors transport the same inherited ordinary triangle; probes create no new triangle. This is the finite mechanism that extends the bounded theorem from rigid supports to arbitrary complete labelled subdivision words.

7 Representative exact calculations

7.1 Tree-sunlet factorization

For the three-sunlet orientation at leaf three,

$$q_{xyz} = a_x b_y c_z \{ \delta f_y d_z + (1 - \delta) f_x e_z \}, \quad x + y + z = 0.$$

With $X_h = q_{hh0}$, $Y_h = q_{h0h}$, $Z_h = q_{0hh}$, and $V = q_{CTG}$, exact substitution gives

$$V^2 X_g - X_s^2 Y_g Z_g = -a_s^2 b_s^2 a_g b_g c_g^2 f_s^2 \delta (1 - \delta) d_g e_g (1 - f_g)^2.$$

This identity is used on whole semi-directed maps. A rooted restriction type is not used as a topology oracle.

7.2 Bridge normalizer

For an unmarked degree- d component, the pair-anchor rows $(12), (13), (23), (14), \dots, (1d)$ have rank d and leading determinant -2 . If r_{ij} are positive anchor ratios, the incidence scales are

$$a_1 = \sqrt{r_{12} r_{13} / r_{23}}, \quad a_2 = \sqrt{r_{12} r_{23} / r_{13}}, \quad a_3 = \sqrt{r_{13} r_{23} / r_{12}}, \quad a_k = r_{1k} / a_1.$$

The formula is applied independently to the $\{C, T\}$ and G sectors.

7.3 Certificate schema for a direct polynomial

A sparse direct certificate stores

$$P(q) = \sum_m c_m \prod_{j \in m} q_j,$$

with integer coefficients and explicit Fourier-coordinate tuples. Its replay regenerates source and target polynomial maps from the two graph encodings, forms $P \circ \Phi_S$ and $P \circ \Phi_T$ over $\mathbb{Z}$, checks $P \circ \Phi_T = 0$ coefficientwise, checks a nonzero source polynomial hash, and evaluates it at a rational point satisfying every strict K2P and inheritance inequality. Degrees two, three, four, and five are stored and mutated separately so that certificate reassignment is rejected.

8 Full weak-class certificate

The base graphs are encoded by

$$\begin{aligned} W : \quad & rS, rL_0, SU, SV, UX, VZ, ZX, UV, ZL_1, XL_2, & \text{Ret} = \{V, X\}, \\ W' : \quad & rS, rL_0, SU, SX_0, VX_0, UX_1, VX_1, UV, X_0L_1, X_1L_2, & \text{Ret} = \{X_0, X_1\}. \end{aligned}$$

Complete rooting censuses are $(5, 2, 3)$ and $(7, 2, 5)$ for admissible, tree-child, and non-tree-child rootings. Exact mixed-incidence comparison returns relation **none**, including after forgetting head flags on the unique triangle of each graph.

Put $\delta = 2^{-30}$. The complete parameter mapping is:

datum	W	W'
seven internal pairs	$(1/7, 1/7)$	$(1/4, 1/4)$
inheritance 1	$Z \rightarrow X : 15996/16339; U \rightarrow X : 343/16339$	$V \rightarrow X_1 : 1/2; U \rightarrow X_1 : 1/2$
inheritance 2	$S \rightarrow V : 1/8; U \rightarrow V : 7/8$	$V \rightarrow X_0 : 1/6; S \rightarrow X_0 : 5/6$
leaf 0 pair	$(86779\delta/80, 86779\delta/80)$	$(16\delta/3, 16\delta/3)$
leaf 1 pair	$(320\delta/253, 320\delta/253)$	$(32\delta/9, 32\delta/9)$
leaf 2 pair	$(114373\delta/20240, 114373\delta/20240)$	$(96\delta/5, 96\delta/5)$

In character code order $0, C, G, T = 0, 1, 2, 3$, the common coordinate order is

$$(000, 0CC, 0GG, C0C, CC0, CGT, CTG, G0G, GCT, GG0).$$

The value is 1 at 000, δ^2 at the six two-nonzero coordinates, and $4\delta^3/5$ at the three all-nonzero coordinates.

The normalized tensors before pendant scaling are, respectively,

$$\left(1, \frac{64009}{457492}, \frac{64009}{457492}, \frac{6400}{39229939}, \frac{1}{1372}, \frac{4048}{39229939}, \frac{4048}{39229939}, \frac{6400}{39229939}, \frac{4048}{39229939}, \frac{1}{1372}\right)$$

and

$$\left(1, \frac{15}{1024}, \frac{15}{1024}, \frac{5}{512}, \frac{27}{512}, \frac{9}{4096}, \frac{9}{4096}, \frac{5}{512}, \frac{9}{4096}, \frac{27}{512}\right).$$

The nonzero rank-nine minors use parameter columns $0, \dots, 8$. Their rows and determinants are

map	output rows	determinant
W	$(1, 2, 3, 5, 4, 7, 6, 8, 9)$	$10368019213741323/563981315074464023964442388464888915634290688$
W'	$(1, 2, 3, 5, 4, 6, 7, 8, 9)$	$1435825/85002596691653613846528$

For cherry induction use actual physical pairs $(u_s, u_g) = (2/5, 4/9)$ and $(v_s, v_g) = (3/7, 5/11)$. The observable block $(u_s/v_s, u_s v_s, u_g/v_g, u_g v_g)$ has determinant $2464/675$. More explicitly, for an outside leaf j and new cherry leaves a, b ,

$$\frac{Q_{j=C, a=C, b=0}}{Q_{j=C, a=0, b=C}} = \frac{u_s}{v_s}, \quad Q_{a=C, b=C} = u_s v_s,$$

with all omitted characters zero; replacing C by G gives the other two observables. Square-root recovery gives the four edge factors, and division by the corresponding nonzero factor recovers every old tensor coordinate. This is the analytic local inverse; graph pruning separately preserves weak-not-strong status and inequivalence.

9 Theorem-to-artifact crosswalk

All paths below are relative to the `k2p_level2_identifiability_closure` project root. “Primary” is the producer/certificate path; “replay” is a separately implemented checker or audit. Historical files not named here are not promotion evidence. This printed table is a reader map, not the complete machine schema. The field-rich authority is `proof_compression_submission/crosswalk/THEOREM_ARTIFACT_CROSSWALK.json`, which records each claim’s declared artifact schemas, exact byte hashes, producer, replay and mutation artifacts, environment profile, and whether a runtime is known or deliberately null.

Executable entry points are invoked from the project root with the pinned Python environment shown in Section 11.

claim	primary proof/certificate	independent replay or mutation
K2P domain, subdivision, rooting	work/domain_rooting_closure/PROOF.md; work/domain_rooting_closure/domain_rooting_certificate.json	work/domain_rooting_closure/verify_domain_rooting.py
Quartet signs and tree of blobs	work/quartet_separation_closure/PROOF.md; work/quartet_separation_closure/quartet_logic_certificate.json	work/quartet_separation_closure/verify_quartet_logic.py; work/adversarial_proof_review/verify_topology_direction.py
Whole-map $\mathcal{T}_i$ and triangle germ	package/original/checkpoint_2/continuation_2/K2P_TREE_SUNLET_SIGN_CERTIFICATE.md; package/original/checkpoint_2/continuation_2/K2P_TRIANGLE_GERM_EXACT.md	package/original/checkpoint_2/continuation_2/verify_triangle_and_sunlet.py; work/final_theorem_release/no_assert_triangle_sunlet.py
Bridge fibre, marginal, gluing	work/bridge_marginal_closure/PROOF.md; work/bridge_marginal_closure/certificate.json	work/bridge_marginal_closure/verify_bridge_marginal.py; work/adversarial_proof_review/PHYSICAL_LOCAL_PRODUCT_REPAIR.md; work/adversarial_proof_review/verify_adversarial.py
Global classification, genericity, reconstruction, and continuous time	work/global_theorem_closure/promotion_manuscript/K2P_SAME_PROMOTION_MANUSCRIPT.md	work/global_theorem_closure/promotion_manuscript/QUANTIFIER_AUDIT.md; work/global_proof_adversary/AUDIT.md
Four-port raw universe and terminals	work/corrected_composite_ledgers/artifacts/raw4_corrected_composite_ledger.jsonl.gz; work/corrected_composite_ledgers/artifacts/raw4_terminal_certificate_registry.json.gz	work/corrected_composite_ledgers/verify_corrected_composites_independent.py; work/corrected_composite_ledgers/run_composite_mutations.py
Five-port θ_2	work/corrected_composite_ledgers/artifacts/theta2_corrected_composite_ledger.jsonl.gz	work/corrected_composite_ledgers/verify_corrected_composites_independent.py; work/corrected_composite_ledgers/run_composite_mutations.py, family theta2

claim	primary proof/certificate	independent replay or mutation
997-parent forest	work/restoration_sign_reclassification/corrected_restoration_forest.json	work/restoration_sign_reclassification/corrected_restoration_replay_certificate.json; work/restoration_sign_reclassification/verify_corrected_restoration_forest.py; work/restoration_sign_reclassification/mutate_corrected_restoration_forest.py
Three-port cycle closure	work/cycle_three_port_closure/PROOF.md; work/cycle_three_port_closure/promotion/cycle_promotion_certificate.json	work/adversarial_proof_review/verify_corrected_cycle_promotion.py; work/adversarial_proof_review/test_cycle_promotion_mutations.py
Probe input completeness	work/adversarial_proof_review/PROBE_INPUT_CONTRACT.md; work/adversarial_proof_review/probe_input_contract.json	work/adversarial_proof_review/verify_probe_input_contract.py; work/adversarial_proof_review/test_probe_input_mutations.py
One-/two-port probe closure	work/probe_coherence_corrected/PROOF.md; work/probe_coherence_corrected/probe_coherence_certificate.json	work/probe_coherence_corrected/independent_verification.json; work/probe_coherence_corrected/verify_probe_coherence_corrected.py; work/probe_coherence_corrected/run_probe_coherence_mutations.py
Primitive probe audit	work/global_proof_adversary/probe_full_audit/independent_probe_graph_audit_certificate.json	work/global_proof_adversary/probe_full_audit/independent_probe_graph_audit.py; work/global_proof_adversary/probe_full_audit/independent_probe_mutation_report.json
Weak sharpness	work/weak_sharpness_closure/PROOF.md; work/weak_sharpness_closure/weak_sharpness_certificate.json	work/weak_sharpness_audit/PROOF_AUDIT.md; work/weak_sharpness_audit/certificate.json; work/weak_sharpness_audit/test_mutations.py

claim	primary proof/certificate	independent replay or mutation
Unified theorem lock	<code>work/final_theorem_release/RELEASE_LOCK.json;</code> <code>work/final_theorem_release/verify_final_theorem_release.py</code>	<code>work/final_theorem_release/corrected_universe_independent_replay.json;</code> <code>work/final_theorem_release/run_release_mutations.py</code>

10 Frozen hash anchors

The following SHA-256 values are the compact anchors most useful for checking that the intended finite universe has been unpacked. The outer release lock authenticates all nested files and should remain the authority if any table is updated.

artifact	file SHA-256
unified release lock	0c17eeaa3344f0982998ea694c1eb92f72f5ced0841e2acad0d39566e2ec71c3
raw-four composite ledger	431dac8898ad2a724d12c200687de1b377723e302214a79a11a03524a4084b96
raw-four independent replay	3e067178ca56f94672151dcf42d48e3b05b5cec411cd709c8b1ae00faf20b620
raw-four terminal registry	0a1818655429d60660c1ed87f3fbe412701f386b081562b3a4caa54079069f1d
raw-four mutation report	2e5f23694ca89b1a021dfc58af829b6596cd497ec7469ca2f038095699b1e672
five-port θ_2 composite ledger	4cbd7b774adccaafc81338ce9093e33f4abcae8d75664c9d4c9ecc582a80cc58
five-port independent replay	58c37895cc896c69ba1c25db1476f83cd22dd2e2986535b358222698de94cc42
corrected 997-parent forest	43bd2be5e7626a954fc4fa4cf45e8d0e6483c947ddc9cba80f2b1a13351bc3a8
restoration independent replay	24fa2e61f60610a8b24c4107ec7f866278f0cc671ca203d7aaa40a37bea291dd
probe input contract	7f686ae99dd5e6dafc1c04396b711d294a0bddd6a25574f9ea809b831ad7b377
full probe primary	6c228e0d495fbc68b1df4ddb571558425a35e081cc22b954f232f8a3458991d3
full probe independent graph audit	879a397a165e19c162561435e7996a5de5ebdf5e9a7dc38af61a13cc6aa36410
full probe adversarial mutation report	b0df0584163150c9a823b4e364b8ee46c196ae8abb28fdca4d3d5893a97bfea7
weak-sharpness primary	e66c78a0aeab990b4dc448f4f064b37e1e15ecbff75a5f472bf116d4464378bd
weak-sharpness independent audit	cfd8d3a2ebc7431d141cac6ebe943e25730eb086fbc84b52833a40bee40a5d52

The combined one-/two-port ordered-ledger root is

7868fed6f8e0c10fcb9740da8ffdc7b7f64ea68939c99cba6f364da4cfd90bf50.

11 Replay protocol

Run from the project root. The locked environment uses Python 3.10 or newer; the qualified release used Python 3.14.6, NetworkX 3.5, and SymPy 1.14.0.

```
python3 -m venv .venv
.venv/bin/python -m pip install --upgrade pip
.venv/bin/python -m pip install \
-r work/final_theorem_release/requirements.txt
```

```
.venv/bin/python -B \
  work/final_theorem_release/build_release_lock.py --check --require-ready
.venv/bin/python -B \
  work/final_theorem_release/verify_final_theorem_release.py --quick
.venv/bin/python -B \
  work/final_theorem_release/run_release_mutations.py
```

For full primitive regeneration:

```
.venv/bin/python -B \
  work/final_theorem_release/verify_final_theorem_release.py --full
```

The entry points reject optimized Python (`python -O`). Full mode regenerates large ledgers by streaming; it does not require an uncompressed multi-gigabyte temporary JSONL file. Before a public submission snapshot, a clean full replay should be run against the final manuscript/package binding and its wall time and peak memory recorded. Those final performance values and the deterministic archive SHA-256 are pending that run.

12 Mutation coverage

The independent suites reject at least the following semantic faults:

- omitted or duplicated raw relation;
- swapped or false topology reason;
- false target rank upper bound or rank exclusion;
- reassignment of a terminal class;
- omitted restoration role or child;
- wrong source insertion, target attachment, or parent;
- restoration cycle or missing continuation layer;
- broken boundary transport or parent restriction;
- omitted probe anchor, pendant site, reticulation-incoming site, or root-suppressed site;
- splitting the two artificial root halves into distinct physical sites;
- source–target reversal omission;
- reassignment of quadratic, cubic, quartic, quintic, or $\mathcal{T}_i$ certificate;
- reintroduction of a rooted `triple_type` topology oracle;
- changed sharpness arrows, parents, inheritance complements, Fourier coordinates, rank minors, or incomplete rooting list; and
- optimized-Python execution that would disable fail-closed checks.

Mutation rejection demonstrates that the verifier watches these data. It is not a substitute for the direct mathematical implication associated with each certificate type.

13 Recent-source verification note

Citation metadata were checked on 21 August 2026 against primary publisher or authoritative preprint records where available:

source	verified locator and status
Kimura (1980) [9]	Springer version of record, https://doi.org/10.1007/BF01731581 , vol. 16, pp. 111–120.
Evans–Speed (1993) [5]	Project Euclid/Annals record, https://doi.org/10.1214/aos/1176349030 , vol. 21, pp. 355–377.
Sturmfels–Sullivant (2005) [10]	author/arXiv and journal metadata agree on J. Comput. Biol. 12(2), pp. 204–228, https://doi.org/10.1089/cmb.2005.12.204 .
Gross et al. (2021) [6]	Springer version of record, https://doi.org/10.1007/s00285-021-01653-8 , J. Math. Biol. 83:32.
Cox–Gross–Martin (2025) [3]	Springer version of record, https://doi.org/10.1007/s11538-025-01506-1 , Bull. Math. Biol. 87:132.
Ardiyansyah (2021) [1]	arXiv primary record https://arxiv.org/abs/2104.12479v1 ; the abstract states the simple/semisimple level-2 invariant scope used in the related-work summary.
Holtgrete et al. (2025) [7]	Springer version of record, https://doi.org/10.1007/s11538-025-01549-4 , Bull. Math. Biol. 87:168; the publisher records a 14 January 2026 correction.
Holtgrete et al. (2026) [8]	Springer version of record, https://doi.org/10.1007/s12064-025-00453-8 , Theory Biosci. 145:4.
Englander et al. [4]	bioRxiv DOI https://doi.org/10.1101/2025.04.18.649493 ; theorem numbering and the 4 July 2026 version-4 content were checked against the locally archived source XML/PDF.
Brits et al. (2026) [2]	arXiv primary record https://arxiv.org/abs/2607.12919v2 ; version 2 dated 29 July 2026.

The local K2P/K3P theta-collision manuscript is provisional and has no verified public bibliographic status in this draft. It is therefore not cited as a published result or used as theorem evidence.

14 Pending submission metadata and licenses

The following items require explicit human decisions before submission:

- corresponding email address;
- author-contribution statement;
- final funding and competing-interests declarations;
- final article and supplement titles, journal formatting, and submission date;
- paper, data, and code license choices;
- final public repository URL, immutable commit/tag, and archive SHA-256;
- whether and when to create a GitHub/Zenodo release and DOI; and
- final clean full-replay wall time and peak-memory report.

No email address, funding statement, competing-interests declaration, or license has been inferred from older project files. This draft asserts none of them. A DOI-bearing release should be created only for a genuinely final immutable submission package.

Disclosure

Generative AI assisted with discovery, code, drafting, and adversarial review. The cited independent replays are separately implemented programs, not independent human review. The author accepts responsibility for the final mathematics, software, and submission.

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